

# The curled torus burps: activated escape and conversion in a graph model of emergent geometry

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We study activated escape from a curled torus in four-regular bipartite graph energies adjacent to combinatorial quantum gravity (CQG). A coefficient  $\lambda$  multiplies the local square-surplus penalty; only  $\lambda = 1$  is CQG, while our decay experiments use  $\lambda > 1$ . The ideal curled torus lies exactly  $4(\lambda - 1)$  per vertex above zero-energy states. For a specified Metropolis edge-switch chain, enumerated moves predict mean first-exit times without fitted parameters, consistently with direct measurements. During conversion, most vertices retain curled-like or flat-like square-count signatures, and a connected flat-like region usually dominates at half conversion; multiple early patches occur. Topology checks certify tori for all 120 saved endpoints at  $\lambda = 1.05$ , without establishing macroscopic period growth. At  $\lambda = 1.25$ , energy-conserving graph-plus-reservoir runs escape with a 12-unit budget over  $N = 48$ –192; exhaustive neutral-class enumeration certifies the matching lower bound at those sizes. The product depends on the reservoir’s thermal response; full release requires a zero-energy endpoint. Detected-decay waits are a distinct observable with unresolved long tails: a dedicated registered tail test fails in two cells. All three registered conversion maps remain inconclusive. These results support a bounded kinetic claim while leaving the conversion mechanism and reservoir equilibration open.

## I. INTRODUCTION

In combinatorial quantum gravity (CQG) [1–3], a discrete Ricci curvature favors graph configurations with geometric structure. Published equilibrium studies address the emergence of geometry from a random phase. Their ensembles require care: Ref. [2], Sec. 4, describes a restricted bipartite simulation in which no edge has surplus squares, whereas Ref. [3] discusses the uncapped curvature action. Ref. [4] finds first-order behavior in a related global-only model without the same fixed bipartite ensemble. These are context, not identical-model validation. Kelly’s thesis [5], p. 219 and Fig. 4.58, reports small four-regular hysteresis loops over  $N = 50$ –500 but declines a definitive first-order inference.

Here we ask how a specified ordered precursor escapes and converts at fixed coupling. The precursor is a torus with a four-edge short period. We call its energy-releasing conversion a “burp.” We use “decompactification” only for removal of that four-cycle signature: a macroscopic increase in both independent geometric periods has not been measured. For example, a  $6 \times L$  torus already has the flat local signature while retaining a fixed short period. The work concerns a deformation next to CQG, not a CQG allotrope in the sense of Ref. [6].

Why deform CQG at all? At  $\lambda = 1$  an edge carrying three squares costs exactly what an edge carrying two costs, so the flat torus, the curled torus and the 4-cube are degenerate: conversion between these ideal states releases no energy. Equal energies need not imply equal free energies. The coefficient  $\lambda$  lifts that degeneracy and changes nothing else: the states, the moves and the hard-

core rule are those of CQG. For  $\lambda > 1$  the ground states are the arrangements with two squares on every edge; this condition alone does not specify connectedness or topology. Each curled direction costs a known energy per vertex. The family is therefore a controlled laboratory for a change between two ordered arrangements, in which the barrier, the release and the relic can each be counted exactly, and  $\lambda \rightarrow 1$  connects it to CQG.

Every kinetic statement below, about waiting times, barriers and fronts, concerns one Markov chain: the local edge switch used to sample the model [2], with Metropolis acceptance and an ordered-pair proposal that are our choices (Ref. [3] uses Glauber acceptance). Detailed balance gives the Boltzmann stationary measure, but does not establish mixing at production sizes; we do not claim physical time evolution of emergent space. What does not depend on it is the energetics: the gap between the two arrangements, the cost of the cheapest single move out, and hence the least energy a sealed system needs to leave by such moves. What does depend on it are the rates, including how the waiting time scales with size (Sec. II).

## II. MODEL AND EXACT RESULTS

States are simple undirected four-regular graphs on a fixed labelled bipartition, with  $N/2$  vertices on each side, obeying the hard-core rule (no two vertices share more than two neighbours) [2, 3]. Connectedness is not enforced. At fixed dimensionless coupling  $g > 0$  the target is  $\pi(G) \propto e^{-H(G)/g}$ ; a valid symmetric proposal is accepted with probability  $\min(1, e^{-\Delta H/g})$ . Energies and  $g$  are in the same dimensionless units. With  $S$  the number of squares,  $S_e$  the squares on edge  $e$  and

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$$X = \sum_e (S_e - 2)_+,$$

$$H = 16(N - S) + 4\lambda X = 4 \sum_e [(2 - S_e) + \lambda(S_e - 2)_+]. \quad (1)$$

At  $\lambda = 1$ , the CQG convention is  $H = -4 \sum_v \sum_{w \sim v} \kappa(v, w)$  [3], Eq. (21). For these bipartite graphs  $\kappa(v, w) = -(2 - S_{vw})_+/2$ , giving  $H = 4 \sum_e (2 - S_e)_+$  with each undirected edge counted once. For  $\lambda \neq 1$ ,  $H$  is not this curvature action. The hard-core rule limits  $S_e \leq 3$ .

*Ground state (exact).* Rewriting Eq. (1) as

$$H = 4 \sum_e [(2 - S_e)_+ + (\lambda - 1)(S_e - 2)_+]$$

shows  $H \geq 0$  for  $\lambda \geq 1$ . At  $\lambda > 1$ ,  $H = 0$  if and only if every edge has two squares. At  $\lambda = 1$ , every edge need only have at least two. Flat lattice tori are ground states; disconnected unions are allowed too. Topology must be checked separately.

*Ladder (exact).* Periodic bipartite tori require even periods. For even  $L > 4$ , the  $4 \times L$  torus has  $S = 5N/4$  and  $X = N$  (the edges of each wrap-around 4-cycle carry three squares), so  $H/N = 4(\lambda - 1) \equiv \varepsilon$ , its excess above a zero-energy state; the 4-cube has  $H/N = 8(\lambda - 1)$ . Each curled direction costs  $4(\lambda - 1)$  per vertex in these ideal products (Fig. 1);  $4 \times 4$  is the 4-cube, not the one-curved case. A decay with residual graph energy  $H_f$  releases  $Q = \varepsilon N - H_f$ , not necessarily  $\varepsilon N$ .

*Counting signatures.* Define  $d(v)$  as the number of pairs of incident edges that close no square: 2 on a flat lattice torus, 1 on a one-curved torus, 0 on the 4-cube. This is not a geometric dimension. Under the hard-core rule each edge pair closes at most one square. If  $q_v$  is the number of squares through  $v$ , then  $d(v) = 6 - q_v$ ,  $\sum_v d(v) = 6N - 4S$ , and the recorded conversion coordinate is

$$f = \frac{5N/4 - S}{N/4} = \bar{d} - 1.$$

Only in a mixture exclusively of  $d = 1$  and  $d = 2$  signatures does  $f$  equal the  $d = 2$  fraction. Melting can raise  $f$  too. Even  $d = 2$  does not certify flat order: a valid single switch at  $N = 64$  produces vertices whose incident edge square counts are  $[1, 2, 2, 3]$ , rather than  $[2, 2, 2, 2]$ . We therefore call  $d = 1, 2$  curled-like and flat-like counting signatures.

*Strict local minima at  $N = 18$  (exact enumeration).* All 26 graph classes are enumerated. Solving the affine inequalities  $H > 0$  and  $\Delta H > 0$  for every valid single switch, rather than scanning a grid, gives precisely two above-ground strict minima on  $\lambda \geq 1$ :

|     |     |                            |
|-----|-----|----------------------------|
| $S$ | $X$ | open interval in $\lambda$ |
| 21  | 12  | (1, 8/5)                   |
| 22  | 16  | (1, 4/3)                   |

This small-system certificate is not a large- $N$  phase diagram.

*The way out.* The Markov chain swaps one partner each between two vertices of the same side (an edge switch preserving the bipartition; Metropolis; the hard-core rule is enforced). A proposal draws two edges one after the other, each uniformly from the  $2N$  edges (a vertex of one side and one of its four slots), so there are  $4N^2$  equally likely ordered proposals, and the same two edges drawn in the opposite order give the same switch: every switch is proposed in two ways. A sweep is  $2N$  attempts. From the perfect  $4 \times L$  torus at the sizes enumerated below, the cheapest switches are of two kinds: move A ( $\Delta S = -2$ ,  $\Delta X = -4$ ; cost  $32 - 16\lambda$ ;  $3N$  distinct switches, hence  $6N$  ordered proposals) and move B ( $\Delta S = -4$ ,  $\Delta X = -10$ ; cost  $64 - 40\lambda$ ;  $2N$  switches,  $4N$  proposals). Move A is therefore offered  $2N \times 6N / 4N^2 = 3$  times per sweep and B twice, independent of  $N$ . Switches that change neither  $S$  nor  $X$  ( $N/2$  of them) do not leave this energy: each reconnects two neighbouring columns at opposite points, twisting the torus globally. Breadth-first enumeration exhausts the reachable neutral classes at  $\lambda = 1.25$  for  $N = 48, 64, 96, 144, 192, 288$ . At each size there are three side-preserving isomorphism classes; all have the same full move census and minimum exit cost 12. Every valid switch is checked at each representative, and every neutral target belongs to the exhausted list. Since relabellings map switches bijectively, this certifies closure at these sizes, not for arbitrary  $L$ . The following two-channel approximation to the mean first-exit time uses counted attempt frequencies:

$$\tau_{AB}(\lambda, g) = [3e^{-(32-16\lambda)/g} + 2e^{-(64-40\lambda)/g}]^{-1}, \quad (2)$$

with A the cheaper for  $\lambda < 4/3$  and both costs positive in the kinetic range used here. For a constant full per-attempt hazard  $p$ , the exact first-exit law is geometric, with mean  $1/(2Np)$  sweeps; its continuous exponential approximation applies for small  $p$ . Every other exit costs at least 10 more than the cheaper of A and B for  $1.05 \leq \lambda \leq 1.45$ , and Eq. (2) leaves them out. Those that break two distant edges at once are offered a number of times per sweep that grows in proportion to  $N$  (87 at  $N = 192$ ); counted exactly, they add at most 2% to the exit rate at  $g = 1.5$  for the  $g = 1.5$  conditions of the decay maps, and 16% at the hottest point of Fig. 2(a) ( $g = 2.5$ ,  $N = 144$ ), within its error bar. That Eq. (2) does not depend on  $N$  is a property of the proposal, not of the barrier: the chain draws its two edges from the whole graph, so any one region is offered a way out about  $1/N$  as often per sweep, and a dynamics that updated every region at its own fixed rate would give a waiting time falling as  $1/N$ .

### III. DECAY AT FIXED COUPLING

Table I defines the distinct stopping times and observables used below.

At  $\lambda = 1.25$  the minimum energetic barrier to leave the perfect curled torus, the cost of move A, is  $\Delta E^\ddagger = 12$ .

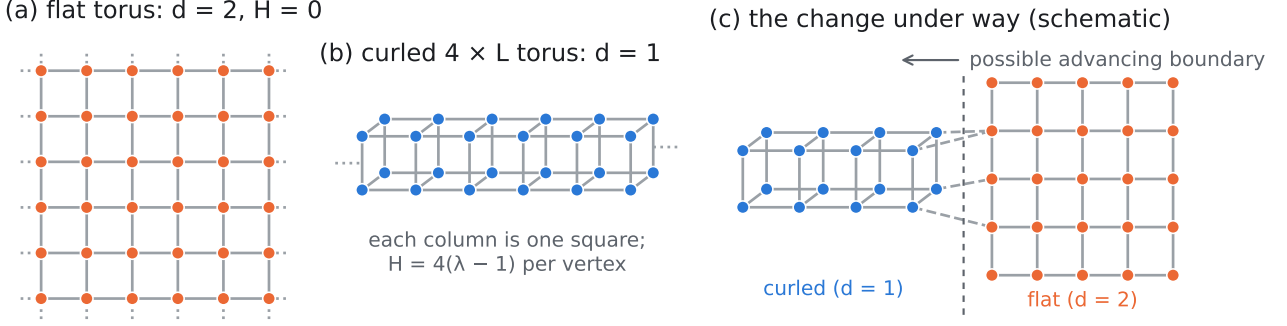


FIG. 1. (a) The flat lattice torus. (b) The  $4 \times L$  torus drawn as a tube: its short direction is a single square, so the edges round each column carry three squares. (c) Schematic of the change of Sec. III, not a simulation snapshot: a possible boundary between curled and flat regions. Neither the schematic nor the square counts establish the number of moving interfaces. The graph ensemble also permits disconnected states.

TABLE I. Observables and clocks. Counts and components are measured on the evolving graph. The terms “first exit,” “detected decay” and “conversion” are not interchangeable.

| Observable           | Operational definition                                                                                                                                      | Prediction or limitation                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| First exit $T_1$     | First accepted move out of $S = 5N/4$ , $X = N$ ; T22 counts each attempt. The exploratory Arrhenius run samples at sweep ends.                             | Eq. (2) is the A+B approximation; full constant hazard gives a geometric law in attempts.                      |
| Successful departure | Last departure before the commitment event $f \geq 0.25$ ; T22 counts exits and returns until that event or its 100,000-sweep cap.                          | 239/240 reach commitment. Reciprocal mean exit count is a conditional empirical success statistic.             |
| Detected decay $W$   | First five-sweep check after 200 with $S/N < 1.25 - 3 \max(\sigma_{\text{rest}}, 10^{-6}) - 10^{-9}$ ; $\sigma_{\text{rest}}$ uses the first forty samples. | Earliest value 205; threshold is trajectory-dependent. The floor curve alone does not reproduce this detector. |
| Conversion snapshot  | First sampled $f \geq 0.25, 0.50, 0.75$ after the resting window, where $f = 5 - 4S/N$ .                                                                    | Histograms of $d$ and induced $d = 2$ components refine the square count; neither certifies a geometric front. |
| Energy release       | $Q = H_i - H_f$ for a saved endpoint; separately, the CSV release is $H_i/N$ minus the final window mean of $H/N$ .                                         | Full ideal decrease requires $H_f = 0$ ; a window mean and a saved configuration need not agree.               |
| Persistence          | More than half a cell has $f_{200} < 0.25$ .                                                                                                                | A finite-duration criterion at $g = 1.5$ , not an energetic spinodal.                                          |

This is a first-exit cost, not a proven minimum maximum energy along a complete conversion path. Whether an excursion grows or returns is a separate question. In an exploratory test (the prediction was written into the run’s configuration before it ran, but not pre-registered), over twelve conditions ( $g = 1.4$  to  $2.5$ ,  $N = 64$  and  $144$ , 16 runs each), the measured mean waiting time against the move-A term of Eq. (2) has a plain mean ratio of 0.98, every ratio between 0.71 and 1.62, over a ninety-fold range of times [Fig. 2(a)]. A weighted straight-line fit of  $\ln \tau$  against  $1/g$  gives effective slopes of  $14.4 \pm 1.0$  at  $N = 64$  and  $12.1 \pm 0.9$  at  $N = 144$ . The first differs from 12 by 2.4 quoted standard errors. Multiple channels give curvature in this plot, so these slopes are not independent measurements of a collective barrier. Inverse-

variance weighting based on 16 skewed waits can favor shorter sample means; we use the direct first-exit comparison below as the main rate test. Model-conditional intervals are given in the Supplemental Material. The exploratory scan reuses each replica’s seed across couplings, so its fitted errors and pooled tests do not account for dependence between cells. Including move B lowers the predicted waiting time by 15% at  $g = 1.5$ ; across  $\lambda$ , the full Eq. (2) is compared separately with first exits and detected-decay waits below.

The decay itself was pre-registered: at  $\lambda = 1.25$ ,  $g = 1.5$ , thirty decays at each of  $N = 64, 96, 144, 192$ , and the adjacency read when the conversion fraction first crossed 25, 50 and 75%. The predictions were a memoryless waiting time (coefficient of variation between 0.7

and 1.3), at least 80% of vertices at  $d \in \{1, 2\}$  at half conversion, and the converted vertices in one piece holding at least 70% of them, averaged across replicas. All three component tests hold at every size and on two independent sets of seeds (Table II). They show dominant connected flat-like regions in the evolving graph at half conversion, not unique nucleation or a single moving interface. The measured means show no clear decrease with  $N$  over these sizes; detected waits need not have the same size dependence as first exits. The same pre-registration listed  $\lambda = 1.5$ ; there the torus does not persist under the tested protocol at  $g = 1.5$  (its number of squares per vertex falls to 0.95, below the flat torus's 1, within 50 sweeps in every replica), and no verdict was applied. A separate pre-registered test of whether a coarse law of a single front governs these decays (T12; 64 decays at  $g = 1.5$  and 2.0, under Metropolis and Glauber acceptance) came out NOT ESTABLISHED: 21 of the 59 that nucleated converted in more than one patch, mostly at the warmer coupling, and the front's rate fell roughly as  $N^{-0.7}$ .

The coefficient of variation is a coarse diagnostic, not sufficient to establish memorylessness. In an exploratory distribution analysis, first-exit data are consistent with an exponential approximation [Fig. 2(c)]: pooled over conditions after dividing each time by its condition's mean, the 432 first exits we recorded give a Kolmogorov–Smirnov distance  $D = 0.035$  from the unit exponential (nominal  $p = 0.53$ ; shared-stream caveat in Methods), and the 240 of them counted move by move (Sec. V) are consistent with an exponential using the mean of Eq. (2), nothing estimated ( $p = 0.85$ ). This does not establish exponential tails in each cell. Detected-decay waits show stronger departures, including the later T38 tail-test failures (Sec. V).

The ideal-state energy difference is exact. After settling, 24 to 28 of 30 decays per size have a window-averaged release within 1% of  $\varepsilon$  per vertex. This energy criterion alone does not certify torus topology. Other endpoints contain defects, read from their wiring: most often combinations of small defects of 8 to 45 units, and less often the four-point relic of Sec. IV, a closed loop of

four vertices at  $d = 1$  with  $S = N + 1$  and, in its commonest form,  $X = 6$ , costing  $24\lambda - 16$  (14 at  $\lambda = 1.25$ ). Whether these states are stable or only slow is not known. The energy criteria were developed after inspecting endpoints, so this energy analysis is exploratory. Under the resulting criteria, the full two-state classification holds at  $N = 64, 96, 192$ , while one  $N = 144$  run fails because its state changes inside the final measuring window. The prespecified waiting and morphology criteria pass at all four sizes in both sets of thirty runs. A passing component test is not a passing combined classification.

A separate equilibrium scan at  $\lambda = 1.25$  has a single energy-distribution peak over the tested sizes and couplings, with an INCONCLUSIVE classification under its stated rules. Neither that finite-size observation nor the kinetic conversion studied here establishes a thermodynamic first-order transition.

#### IV. SEALED SYSTEM

Giving the graph a Creutz reservoir [7] replaces fixed  $g$  by energy-conserving dynamics on graph plus stores. Each valid candidate selects one of  $C$  nonnegative stores uniformly; the move is refused if that store cannot pay  $\Delta H$ , and otherwise its energy changes by  $-\Delta H$ . The stores and edge proposals are globally selected: the seed is a total activation budget, not energy deposited at a chosen vertex. The conserved quantity is  $E_{\text{tot}} = H + \sum_i E_i$ . The recorded maximum drift is zero in all T9 runs at  $\lambda = 1.25$ ; this is an observed numerical check, not a claim about bitwise arithmetic at arbitrary  $\lambda$ .

*Thermometer and ensemble.* At  $\lambda = 1.25$ ,  $\Delta H = -16\Delta S + 5\Delta X$  is integral. The initial T9 stores contain 0 or 12. Allowed exit costs include 12 and 25, with greatest common divisor one, so there is no common coarser increment than one; accessibility of all allocations is not thereby proved. If equilibrated with an approximately canonical graph, an unbounded store with levels  $k\delta$  has a geometric energy distribution, giving

$$u_D(g) = \frac{\delta}{e^{\delta/g} - 1}, \quad g = \frac{\delta}{\ln(1 + \delta/u_D)}.$$

For  $\delta = 1$ ,  $u_D(3.5) = 3.024$ , and  $u_D = 0.5$  corresponds to  $g = 0.910$ . The CSV field `bath_T` is the mean store energy, not this inferred coupling. We report the inversion only as an equilibrium-conditional proxy. Store histograms were not saved, so their canonical distribution cannot be checked from these files.

Symmetric reverse moves support a uniform invariant measure on accessible extended states at fixed total energy. They do not demonstrate equilibration. If all integer bath allocations at energy  $M$  are accessible, their number is  $\Omega_C(M) = \binom{M+C-1}{C-1}$ , where  $M = E_{\text{tot}} - H(G) \geq 0$ . The graph marginal is weighted by this bath multiplicity; changing  $C$  changes the entropy as well as the thermal response.

TABLE II. The decay of the  $4 \times L$  torus at  $\lambda = 1.25$ ,  $g = 1.5$  (first run / independent rerun, thirty decays each). “Largest piece” is the mean fraction of  $d = 2$  vertices in their largest induced component. These are counting observables. Replica distributions and 95% bootstrap intervals are in the Supplemental Material; all 30 replicas per rerun size reach the half-conversion snapshot.

| $N$ | mean wait   | CV          | $d \in \{1, 2\}$ at 50% largest piece |             |  |
|-----|-------------|-------------|---------------------------------------|-------------|--|
| 64  | 871 / 660   | 0.91 / 0.84 | 0.990 / 0.998                         | 0.98 / 0.99 |  |
| 96  | 913 / 928   | 1.10 / 1.02 | 0.990 / 0.988                         | 0.95 / 0.96 |  |
| 144 | 856 / 743   | 1.01 / 0.76 | 0.997 / 0.997                         | 0.92 / 0.96 |  |
| 192 | 1039 / 1428 | 0.87 / 1.19 | 0.995 / 0.995                         | 0.85 / 0.92 |  |

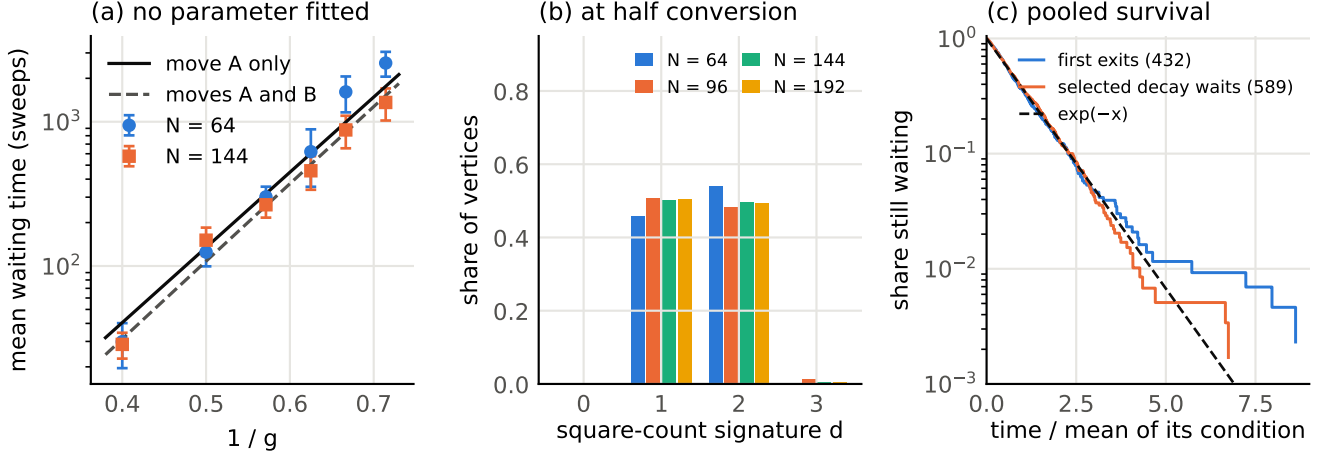


FIG. 2. (a) Mean waiting time of the perfect  $4 \times L$  torus at  $\lambda = 1.25$  against inverse coupling, sixteen runs per point; error bars are one standard error of the mean. Lines: Eq. (2) using move A only (solid) and moves A and B (dashed); nothing is fitted. (b) Share of vertices at each local dimension  $d$  at the first sampled  $f \geq 0.5$ , averaged over the thirty decays per size of the rerun in Table II. (c) Share of tori still waiting against time in units of the mean of its condition, pooled, on a logarithmic scale; the dashed line is the exponential  $e^{-x}$  of a memoryless wait. Blue: all 432 recorded first exits (twelve conditions of panel a, and six cells of the count in Sec. V). Orange: the selected 589 decay waits of the  $\lambda$  map with  $f_{200} = 0$  and  $W > 205$ , shown as  $W - 205$  (Methods). This selection does not establish that no earlier exit occurred. Not pre-registered.

*A size-independent escape budget (exploratory).* With one store initially holding  $s \in \{4, 8, 10, 11, 11.5, 12, 13, 16\}$ , the torus left its initial  $(S, X)$  values in 8 of 8 runs at each tested  $s \geq 12$ , and 0 of 8 at each  $s \leq 11.5$ , for each  $N = 48, 64, 96, 144, 192$  (320 runs of 12,000 sweeps). The runner checks  $(S, X)$  once per sweep: its success flag measures escape, not completed conversion. Eight successes give a two-sided exact 95% binomial interval of  $[0.631, 1]$  at an individual setting; zero of eight gives  $[0, 0.369]$ . The neutral-class certificate in Sec. II proves the lower bound 12 at these sizes: all nonneutral moves cost at least 12, so nonnegative stores with total budget below 12 cannot leave the basin. Observed sufficiency is protocol- and duration-dependent. The escape budget is fixed while the available ideal-state energy decrease  $\varepsilon N$  grows with the system.

*The reservoir’s thermal response.* With the seed in one of  $C$  stores,  $C$  setting the reservoir’s heat capacity ( $N = 64, 96, 192$ ; twenty runs each; 420 runs): the converted region melts into the random phase in 17 to 20 of 20 runs for  $C \leq N/16$  and in 9 to 16 of 20 at  $C = N/8$ ; at  $C = N/4$  most runs end partly converted or partly melted; for  $C \geq N/2$  the “sheet” counting criterion holds in 18 to 20 of 20. Runs last 30,000 sweeps, with  $C = 1, N/16, N/8, N/4, N/2, N, 2N$ . The criterion requires  $f \geq 0.9$ , at least 90% at  $d = 2$ , and less than 5% at  $d = 0$  or  $d \geq 3$ ; it is not a topology test. For perspective, 18/20 and 20/20 successes have exact 95% binomial intervals  $[0.683, 0.988]$  and  $[0.832, 1]$ . The observed crossover lies between  $N/4$  and  $N/2$  on this coarse grid. A conditional equilibrium energy-balance estimate

at crossover coupling  $g_m$  would be

$$C^* \simeq \frac{\varepsilon N + s - U_{\text{graph}}(g_m)}{u_D(g_m)}.$$

Taking  $g_m \simeq 3.5$  and neglecting seed and residual graph energy gives  $N/3.024$ . This is an approximate energy-budget estimate, not a calibrated crossover prediction. Stores have no upper capacity: a small bath heats, it does not fill up. No sustained two-signature stall was observed under the stated criterion and duration; this does not exclude equilibrium coexistence.

*About one relic at every size.* In the coldest box ( $C = 2N$ ), the number of relic defects per converted torus is  $0.90 \pm 0.31$ ,  $0.85 \pm 0.37$ ,  $1.05 \pm 0.22$  and  $0.95 \pm 0.22$  at  $N = 64, 96, 192, 288$  (mean  $\pm$  standard deviation over twenty runs each; a weighted fit against  $N$  gives a slope of  $0.0003 \pm 0.0003$  per vertex, standard error). The defect is the four-point loop above (14 units), its points spanning one to four of the torus’s original columns and lying along the torus, or occasionally a two-vertex twist (20 units). So the count is of order one, with no measurable growth, across the sizes we could reach. A relic here is a connected component of the  $d = 1$  induced subgraph in the final snapshot, classified further where wiring is available. The  $C = 2N$  T10 runs also last 30,000 sweeps. The discrete coupling proxy ranges from 0.84 to 1.01 across their final baths, so even its conditional interpretation does not support a blanket “below  $g = 1$ ” statement. No association is detected between relic position and the start by the specified distance tests: the pre-registered test of whether it sits opposite the start or at it finds neither, and, checked afterwards, a randomisation test that

rotates each relic through every column of its torus with the start held fixed places the observed mean distance at  $p = 0.67$  and  $0.59$  ( $N = 64$  and  $96$ ;  $36$  and  $28$  runs with one relic). These tests can miss other positional structure. The relic count is a finite-duration observation, not a conclusion about the equilibrium density of defects.

## V. DEPENDENCE ON $\lambda$

T8 maps  $\lambda = 1.05$  to  $1.45$  in steps of  $0.05$  at  $g = 1.5$ , using four sizes and thirty runs per cell (Table III); its  $\lambda = 1.25$  comparison uses T7’s data. The registered persistence criterion is that more than half the runs have  $f_{200} < 0.25$ . It holds from  $1.05$  to  $1.35$  at every size, and fails at  $1.40$  and  $1.45$ . This is an operational persistence boundary, not an energetic spinodal: move B still costs energy until  $\lambda = 1.6$ . Equation (2) gives a mean of 200 sweeps near  $\lambda = 1.345$ , but this does not derive the boundary. An exponential first exit with 50% survival at 200 would instead require a mean  $200/\ln 2$ , and  $f_{200}$  also includes growth.

In persistent cells, means at half conversion give 95–100% in the two counting signatures and 76–100% of flat-like vertices in the largest component. At  $\lambda = 1.05$  all 120 saved endpoints have zero energy. Filling every square, checking two faces per edge, a cyclic vertex link, connectedness, orientability and Euler characteristic zero certifies torus topology for all 120 graphs. This does not cover all other datasets or determine period lengths. The full-window-release fraction falls to 79% at  $1.30$  and 46% at  $1.35$ ; the table retains this energy-window criterion rather than treating it as a topology test.

Figure 3(a) compares detected means with the first-exit prediction over two decades; these are distinct observables. For the idealized floor  $W = \max(T, a)$  with exponential  $T$  of mean  $\tau$ ,  $E[W] = a + \tau e^{-a/\tau}$ . At  $a = 205$  this gives 462 and 265 sweeps at  $\lambda = 1.30$  and  $1.35$ , against 403–505 and 280–350 measured. The dashed curve is a recording-floor approximation, not a calibrated detector prediction. Conditioning on actual survival gives the different mean  $E[T | T > a] = a + \tau$ . The detector uses the same trajectory’s first 200 sweeps to estimate its threshold; early conversion changes both the threshold and sample selection (Methods).

T22 counts exits attempt by attempt in forty further runs in each of six cells ( $\lambda = 1.05, 1.10, 1.25$ ;  $N = 64, 96$ ). All 240 first exits are observed; 239 runs reach  $f \geq 0.25$  before the cap. First-exit means are consistent with Eq. (2) in the model-conditional 95% intervals in the Supplemental Material. For completed runs, the reciprocal of mean exits per run is  $0.56$ – $0.68$ ; its replica-bootstrap intervals span roughly  $0.45$ – $0.78$  across cells. This estimates successful departures per counted exit under the  $f \geq 0.25$  commitment rule, excluding the one incomplete run. It is not a derived transmission coefficient. Multiplying the exit rate by this fraction would require independent renewal excursions and control of time spent in

failed excursions and growth.

*Conversion-map diagnostics.* Three independently seeded maps fail their full prespecified acceptance criteria. In the thirty-run map (T8), the wait CV falls outside  $0.7$ – $1.3$  in nine of 24 persistent cells, and the endpoint-energy criterion fails for two to five runs per size at  $\lambda = 1.35$ . In a 120-run map (T23), the sample-size-adjusted exponential CV criterion passes through  $1.25$  and in three sizes at  $1.30$ , but the energy-window criterion fails at all sizes at  $1.30$ . Thermal window averages need not match the energy of a saved endpoint. In a second 120-run map (T24), all required saved graphs are valid, but the CV criterion fails at  $(N, \lambda) = (64, 1.25), (64, 1.30), (192, 1.30), (144, 1.35)$ . The last map tests endpoint validity and reports energy separately; validity is not evidence for full release. All three combined classifications are INCONCLUSIVE. Their counting-signature and dominant-component means pass through  $\lambda = 1.30$ . Full rules and cell results accompany the data.

Multiple early patches are common even when one component dominates later. From  $\lambda = 1.30$  to  $1.35$ , the fraction with several flat-like pieces at quarter conversion rises from  $0.44$  to  $0.69$  in T23 and  $0.43$  to  $0.68$  in T24. The corresponding full-release or zero-energy fractions fall from  $0.71$  to  $0.47$  and  $0.74$  to  $0.49$ , respectively. T24’s “FLAT” class means  $H = 0$  only.

The two exceptional T24 waits at  $N = 64$  are 20,270 sweeps ( $\lambda = 1.25$ ) and 32,155 ( $1.30$ ), or 23.98 and 76.74 times the predicted first-exit means. For 120 exponentials with those means, the bounds  $120e^{-r}$  would be  $4.6 \times 10^{-9}$  and  $5.7 \times 10^{-32}$ . These illustrative null bounds are not final  $p$ -values: the stopping times differ and the cases were examined after observation. They nevertheless preclude dismissing the detected tails as ordinary scatter around that particular first-exit model.

*Dedicated tail test (T38).* A prespecified tail-count test gives the classification TWO POPULATIONS. The rule uses  $w' = W - 200$ ,  $\hat{\tau} = \text{median}(w')/\ln 2$ , and counts  $k_{10}$  above  $10\hat{\tau}$ : TAIL for  $k_{10} \geq 3$ , NO TAIL for  $k_{10} \leq 1$ , otherwise UNCLEAR.

| $\lambda$ | $N$ | recorded / total | $k_{10}$ | result  |
|-----------|-----|------------------|----------|---------|
| 1.25      | 64  | 3942 / 4000      | 2        | UNCLEAR |
| 1.25      | 192 | 1000 / 1000      | 1        | NO TAIL |
| 1.30      | 64  | 3868 / 4000      | 14       | TAIL    |
| 1.30      | 192 | 1000 / 1000      | 5        | TAIL    |

Two TAIL cells trigger the verdict; it does not establish two physical basins. All 190 missing detector crossings nevertheless reach  $f \geq 0.75$ , so they are detector omissions, not simply long survivors. The tail statistic excludes these missing crossings. A descriptive two-exponential fit also includes a fast component near the recording floor; it is not a mechanism diagnosis. No T38 waiting-graph files are present in the frozen **results/**inventory. Endpoints cannot reconstruct their histories. Separating detector effects, failed growth and additional

traps requires the waiting snapshots or new attempt-resolved trajectories with prespecified tail tests.

## VI. DISCUSSION

At  $\lambda = 1$ , which is CQG, the flat, curled and 4-cube tori are degenerate and the change has zero ideal-state energy difference. The burp described here belongs to the  $\lambda > 1$  family; the available decrease  $\varepsilon = 4(\lambda - 1)$  vanishes at the CQG value. The same questions can be asked in CQG itself, at  $\lambda = 1$ , of regions stuck in a different discrete arrangement inside a curved background [6], such as the allotrope of Ref. [3] (Fig. 9): whether such a region is metastable, and whether it gives way by a front with a definite release. The tools used here (exact barrier counting, front tracking, sealed baths) apply unchanged.

The strongest quantitative kinetic result is the agreement of directly counted first exits with a prediction from enumerated moves, with the omitted-channel correction stated. The energetic lower bound for sealed escape is certified over the tested sizes. Conversion snapshots support dominance by a connected flat-like region, but do not establish a unique seed or one moving boundary. Exact endpoint geometry is certified only for the stated T8 set. Detected-decay tails, finite-bath equilibration and period growth remain open; the inconclusive maps and T38 failures are part of the result.

*Labelled or interchangeable vertices.* All runs here, like the published CQG simulations we know of [2, 3], treat vertices as labelled: two graphs that differ only by renaming vertices are different states. Giving each unlabelled class one Boltzmann weight is a different, legitimate modeling choice; quantum indistinguishability alone does not select this classical measure. With

TABLE III. T8 at  $g = 1.5$ . Ranges are cell means over  $N = 64, 96, 144, 192$ , thirty runs per cell;  $\lambda = 1.25$  waits span both T7 sets, its release fraction uses the rerun. Signatures: share at  $d \in \{1, 2\}$  at half conversion. Release: pooled fraction whose final window-averaged release is within 1% of  $\varepsilon$ , not a topology classification. At 1.05, each size has 30/30 and a 95% exact binomial interval [0.884, 1]; other intervals and denominators are in the Supplemental Material. A dash denotes failure of the persistence criterion.

| $\lambda$ | $\tau_{AB}$ | mean wait  | signatures  | release |
|-----------|-------------|------------|-------------|---------|
| 1.05      | 8330        | 9617–11915 | 1.000       | 100%    |
| 1.10      | 4844        | 4154–6626  | 0.998–1.000 | 98%     |
| 1.15      | 2788        | 2191–3834  | 0.996–0.999 | 97%     |
| 1.20      | 1570        | 1132–1716  | 0.996–0.998 | 93%     |
| 1.25      | 845         | 660–1428   | 0.988–0.998 | 88%     |
| 1.30      | 419         | 403–505    | 0.980–0.984 | 79%     |
| 1.35      | 183         | 280–350    | 0.948–0.963 | 46%     |
| 1.40      | 68          | —          | 0.911–0.924 | 31%     |
| 1.45      | 22          | —          | 0.799–0.837 | 7%      |

$n = N/2$  and side-preserving automorphism count  $A(G)$ , a class has  $(n!)^2/A(G)$  labelings. Thus the corresponding target on labelled graphs is  $A(G)e^{-H(G)/g}$ , and the symmetric-proposal acceptance is

$$a(G \rightarrow G') = \min \left[ 1, \frac{A(G')}{A(G)} e^{-\Delta H/g} \right].$$

The ratio belongs inside the minimum; multiplying an already-clipped Metropolis probability is incorrect in general. Every energy and the certified energetic seed lower bound stay unchanged. The perfect curled torus has  $2N$  symmetries and its neutral classes have  $N$  or  $2N$ . The enumerated A-move targets in the  $N = 64, 96$  calculation have two symmetries. For those uphill moves the instantaneous acceptance is suppressed by  $N/2$  to  $N$ . A rate suppression by  $cN$  corresponds to an effective free-energy increment  $g \ln(cN)$ , not a barrier proportional to  $N$ . Neutral-state occupancy, other exit channels and the quotient proposal also affect the lifetime. No full effective rate or decay simulation is supplied for this alternative ensemble.

Not claimed: that this chain is the physical time evolution of anything; a first-order transition in the thermodynamic sense; a latent heat; isotropic decompactification; a unique moving front; a general large-size law beyond the enumerated and measured sizes; or an explanation for the cold-box defect count.

*Outlook (motivation, not a result).* This work is motivated by the hypothesis that space emerged from a specific ordered precursor through a sharp change of this kind, a lattice cousin of false-vacuum decay [8], releasing a definite energy and leaving relics behind. Nothing here tests that hypothesis beyond showing classical activated escape and energy-releasing conversion in a specified graph model. No tunneling or Lorentzian spacetime dynamics has been computed. Geometric periods, interfaces and attempt-resolved commitment histories would make the next tests more discriminating.

## METHODS AND DATA

*Sampler and validation.* Simple bipartite switches preserve degree and the hard-core rule; invalid proposals are rejected. One sweep is  $2N$  attempts. All clocks in this paper use that convention. The stationary measure follows from symmetric proposals and Metropolis detailed balance. Exhaustive small-graph checks and canonical averages validate the sampler at  $N = 16, 18$ ; ergodicity and equilibration at production sizes are not proved. The square-density comparison at  $N = 160$  agrees with Ref. [2], Fig. 8a, to rms 0.005 but differs from Ref. [3], Fig. 3, by up to 0.41 between  $g = 2$  and 6.3. Given the differing stated protocols discussed in the Introduction, this remains an unresolved reproduction issue, not a validation of the entire uncapped equilibrium curve.

*Conversion protocol.* The runner advances in five-sweep blocks, recording square counts and the threshold

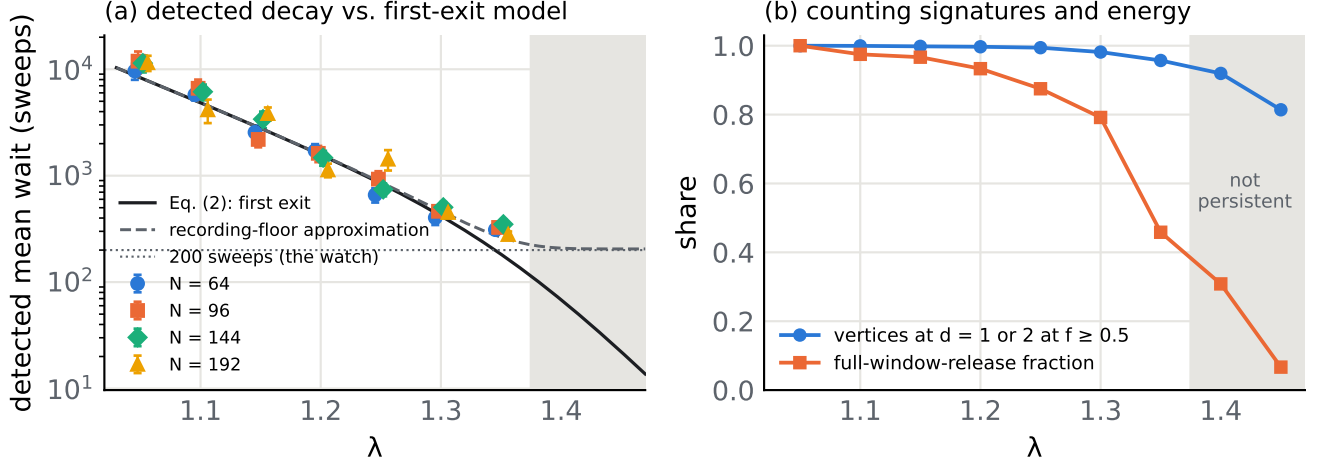


FIG. 3. T8 at  $g = 1.5$ . (a) Detected mean waits in persistent cells; error bars are one standard error. Lines: Eq. (2) for first exits (solid) and the recording-floor approximation  $205 + \tau_{ABE}^{-205/\tau_{AB}}$  (dashed); neither has fitted parameters. Shading marks failure of the operational persistence criterion. (b) Mean share in the two counting signatures at half conversion, and full-window-release fraction, pooled over sizes. Neither series certifies geometry.

crossings in Table I. The main conversion stage stops at  $f \geq 0.98$  or the configured cap; the settling stage follows even if that crossing was not reached. In T7b and T8, settling uses 600-sweep windows. The current stopping rule requires two consecutive window releases to agree within  $0.005\varepsilon$  and the last to lie within  $0.01\varepsilon$  of full release. Otherwise it runs to the configured settle cap. Defected endpoints can therefore exhaust the cap even when long-lived. T7 uses a fixed 300-sweep settling stretch. T7b's nominal main and settle caps are 30,000 sweeps each; T8 uses 100,000 for each. Recorded replay configurations extend selected T7 settling caps. The frozen configuration, not the nominal duration, identifies each replay. Saved final graphs are post-settling states, whereas the 25/50/75% histograms are taken during conversion.

*Analysis design.* Waiting and morphology criteria for the conversion maps, and the dedicated tail-count test, were fixed before their respective runs. Endpoint-energy criteria were partly chosen after inspecting data, so the energy classification is exploratory. The Arrhenius and seed scans, pooled distribution checks, topology certificates and interval estimates are also exploratory analyses, not independent confirmatory experiments. Experiment identifiers such as T8 and T38 connect the paper to the configurations and complete analysis record in the Supplemental Material. Our computational derivations remain unverified by an independent physicist.

*Uncertainty and selection.* Replicas, not vertices, are the independent units within a setting. We report 95% exact binomial intervals for success fractions, percentile bootstrap intervals for replica means (10,000 resamples, seed 20261009), and exponential-model intervals  $[2 \sum T/\chi^2_{2m,0.975}, 2 \sum T/\chi^2_{2m,0.025}]$  for complete first-exit samples. The last are conditional on the contin-

uous exponential approximation, not distribution-free bounds. For the T7b largest-component means, the intervals at  $N = 64, 96, 144, 192$  are respectively  $[0.983, 1]$ ,  $[0.928, 0.986]$ ,  $[0.906, 0.995]$ ,  $[0.864, 0.971]$ . The median is one in each cell, but 0, 1, 3 and 5 of the thirty individual runs have a fraction below 0.7. Thus a passing mean does not describe every replica. Map waiting summaries are restricted to runs reaching  $f \geq 0.75$ ; this selection can favor successful conversions. Counts of observed events, omissions and snapshot availability are given in the Supplemental Material.

*Exploratory distribution checks.* We divide each wait by its cell mean and compare the pooled sample with the unit exponential using the Kolmogorov-Smirnov distance  $D$ . A parametric bootstrap refits cell means in each of 10,000 simulated datasets. It reproduces the sampling grid and residual-time convention, not the full trajectory-dependent threshold. The 432 first exits give  $D = 0.035$ , nominal  $p = 0.53$ ; two of eighteen individual cells have  $p < 0.05$ , both at  $\lambda = 1.05$ . The 240 attempt-counted T22 exits, normalized by Eq. (2) without fitted means, give  $D = 0.039$ ,  $p = 0.85$ . The Arrhenius scan reuses a seed across couplings at each size and replica. The pooled bootstrap simulates cells independently, so its 432-wait  $p$ -value does not model this dependence and is nominal. T22 includes  $\lambda$  in its seed derivation. For detected waits, the unfiltered T7/T8 residual sample (865 waits with  $W > 205$ ) gives  $p = 0.03$ ; T7 alone gives  $p < 10^{-3}$ . Selecting T8 cells passing persistence,  $f_{200} = 0$  and  $W > 205$  gives 589 waits with  $D = 0.030$ ,  $p = 0.55$ . Having  $f_{200} = 0$  does not rule out an earlier exit and return. Pooling and fitting means can mask cell heterogeneity, and KS is not a decisive far-tail test. Three of 200 mixed first-exit and detected waits at  $\lambda = 1.05$  ex-

ceed 75,000 sweeps. Their post-hoc rarity calculation, as well as T24 and T38, motivates an attempt-resolved mechanism audit.

*Data and code.* The source-data snapshot is available at RecreationalPhysics, commit 40b1ae68. The Supplemental Material provides a claim-to-script/configuration/data map, complete check outputs and content hashes in `release_manifest.json`. The release includes the analysis and figure scripts needed to reproduce this paper. Configurations specify seeds and durations; associated metadata record software versions

but do not consistently identify the original execution commit.

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